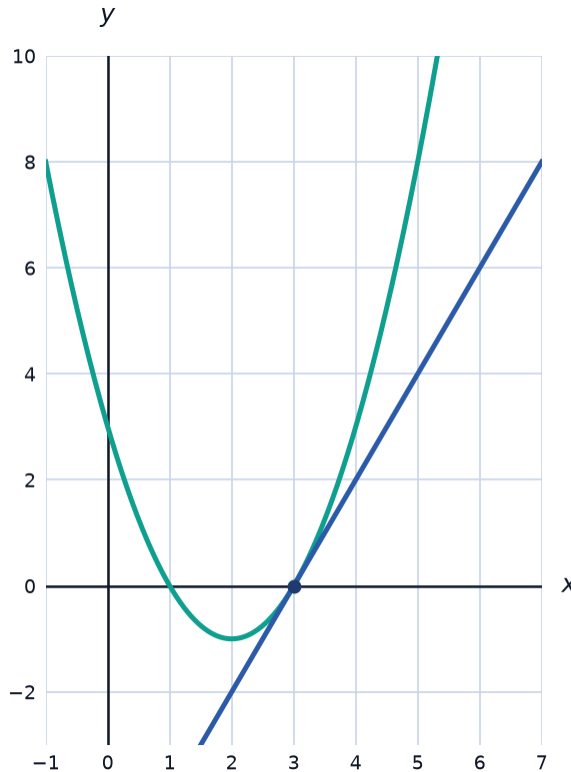


Calculator & Test Strategy



Choosing between graphing and algebra

- 1 A system is given by $y = x^2 - 4x + 3$ and $y = 2x - 6$. Explain how graphing both functions on a calculator can quickly reveal the number of solutions, then find the solutions algebraically.



Using a calculator to check algebraic work

- 2 A student solves $3x - 7 = 2x + 5$ and gets $x = 12$. Verify this solution by substitution, and explain why checking answers this way is a valuable test-taking strategy.

Estimating and eliminating unreasonable answer choices

- 3 A multiple-choice question asks for the value of $\sqrt{50}$, with choices A) 5, B) 7.07, C) 25, D) 100. Without a calculator, use estimation to eliminate unreasonable choices, then find the exact simplified value.

Efficient use of the calculator for systems of equations

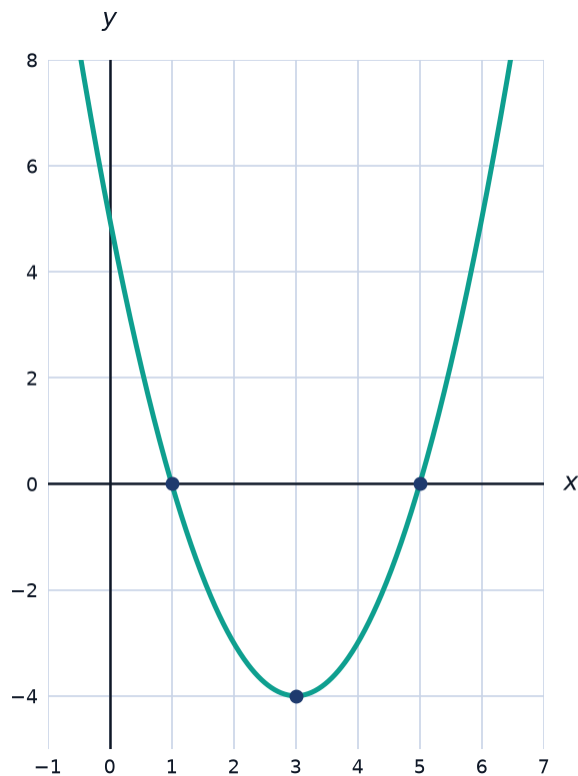
- 4 Explain two different calculator-based strategies (other than the built-in equation solver) for solving the system $2x + 3y = 12$ and $x - y = 1$, and then solve it algebraically to confirm.
-

Time management: identifying quick wins vs. time sinks

- 5 Compare the effort needed to solve $x + 5 = 12$ versus $3x^2 - 2x - 8 = 0$. Explain a time-management strategy for approaching a timed test section with a mix of question difficulties.
-

Using calculator tables to explore function behavior

- 6 A function is given by $f(x) = x^2 - 6x + 5$. Using a table of values at $x = 0, 1, 2, 3, 4, 5, 6$, identify the zeros of the function and the approximate location of its minimum.



Plugging in answer choices strategically

- 7 A multiple-choice question asks to solve $2x^2 - x - 6 = 0$, with choices A) $x = -1.5, 2$, B) $x = 1.5, -2$, C) $x = 3, -2$, D) $x = -1.5, -2$. Test choice C by substitution to determine if it's correct.
-

Substituting numbers for variables in abstract problems

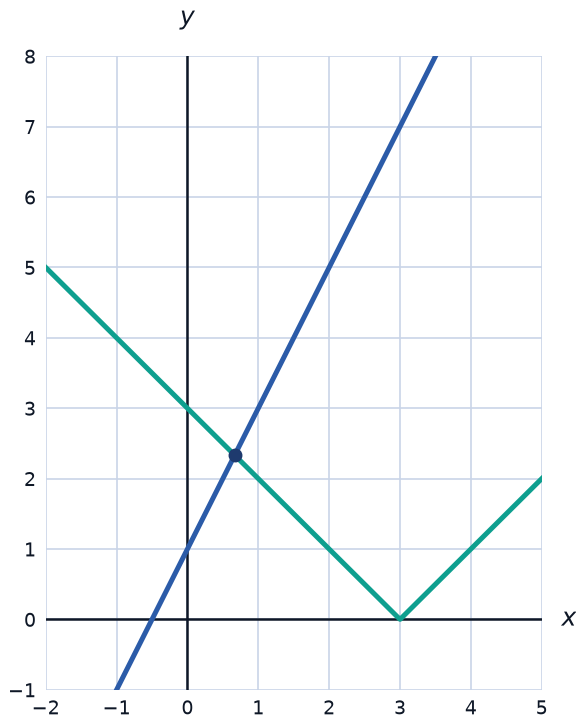
- 8 A problem states: if a is a positive integer, which expression is always even: $2a + 1$, a^2 , or $4a$? Test with $a = 1$ and $a = 2$ to determine the answer.
-

Working backward from answer choices in word problems

- 9 A word problem states that a number increased by 40% equals 84, with answer choices A) 50, B) 60, C) 70, D) 84. Use the answer choices to work backward and find the correct original number.
-

Deciding when a graph is faster than algebra

- 10 A question asks for the number of solutions to $|x - 3| = 2x + 1$ without asking for the actual solution values. Explain how graphing $y = |x - 3|$ and $y = 2x + 1$ answers this faster than solving algebraically, then confirm algebraically.



Recognizing when an exact answer is required vs. an estimate

- 11 A question asks for the exact value of $\sqrt{72}$ in simplest radical form, while a related question asks for a decimal approximation to the nearest tenth. Provide both, and explain why a calculator alone is insufficient for the first part.
-

A multi-part strategy problem: combining estimation, elimination, and verification

- 12** This question has three parts. A word problem states: "A rectangular garden has a perimeter of 46 feet and an area of 120 square feet. Find its dimensions." Answer choices are A) 10 by 12, B) 8 by 15, C) 6 by 17, D) 5 by 18. (a) Use the perimeter condition alone to eliminate choices quickly by estimation. (b) Check the remaining choices against the area condition. (c) State the correct dimensions and verify both conditions algebraically.
-

Avoiding common sign and order-of-operations errors

- 13** A student evaluates -3^2 and gets 9, but the correct answer is -9 . Explain the order-of-operations error, then correctly evaluate -3^2 and $(-3)^2$ to show the difference.
-

Multiple choice — easy

- 14** Without a calculator, estimate $\sqrt{82}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 11
 - b. 7
 - c. 9
 - d. 10
- 15** Without a calculator, estimate $\sqrt{114}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 9
 - b. 10
 - c. 13
 - d. 11

- 16 Without a calculator, estimate $\sqrt{164}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 12
 - b. 11
 - c. 13
 - d. 15
- 17 Without a calculator, estimate $\sqrt{75}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 9
 - b. 7
 - c. 8
 - d. 11
- 18 Without a calculator, estimate $\sqrt{135}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 14
 - b. 11
 - c. 10
 - d. 12
- 19 Without a calculator, estimate $\sqrt{124}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 13
 - b. 12
 - c. 11
 - d. 10

- 20 Without a calculator, estimate $\sqrt{51}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 9
 - b. 6
 - c. 7
 - d. 8
- 21 Without a calculator, estimate $\sqrt{93}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 9
 - b. 12
 - c. 8
 - d. 10
- 22 Without a calculator, estimate $\sqrt{74}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 7
 - b. 9
 - c. 8
 - d. 11
- 23 Without a calculator, estimate $\sqrt{70}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 8
 - b. 10
 - c. 9
 - d. 6

- 24 Without a calculator, estimate $\sqrt{34}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 8
 - b. 5
 - c. 6
 - d. 7
- 25 Without a calculator, estimate $\sqrt{79}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 9
 - b. 8
 - c. 7
 - d. 11
- 26 Without a calculator, estimate $\sqrt{166}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 14
 - b. 15
 - c. 13
 - d. 12
- 27 Without a calculator, estimate $\sqrt{177}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 14
 - b. 11
 - c. 13
 - d. 15

- 28 Without a calculator, estimate $\sqrt{147}$ to the nearest whole number by identifying the two consecutive perfect squares it lies between.
- a. 11
 - b. 13
 - c. 12
 - d. 14
-

Multiple choice — medium

- 29 A number increased by 45% equals 290. Work backward from the answer choices to find the original number.
- a. 290
 - b. 245
 - c. 155
 - d. 200
- 30 A number increased by 80% equals 72. Work backward from the answer choices to find the original number.
- a. 40
 - b. 1
 - c. 72
 - d. 120

- 31** A number increased by 85% equals 296. Work backward from the answer choices to find the original number.
- a. 296
 - b. 245
 - c. 75
 - d. 160
- 32** A number increased by 10% equals 66. Work backward from the answer choices to find the original number.
- a. 60
 - b. 66
 - c. 50
 - d. 70
- 33** A number increased by 5% equals 84. Work backward from the answer choices to find the original number.
- a. 75
 - b. 80
 - c. 84
 - d. 85
- 34** A number increased by 35% equals 270. Work backward from the answer choices to find the original number.
- a. 165
 - b. 235
 - c. 270
 - d. 200

- 35** A number increased by 40% equals 392. Work backward from the answer choices to find the original number.
- a. 320
 - b. 280
 - c. 240
 - d. 392
- 36** A number increased by 55% equals 62. Work backward from the answer choices to find the original number.
- a. 95
 - b. 40
 - c. 1
 - d. 62
- 37** A number increased by 50% equals 330. Work backward from the answer choices to find the original number.
- a. 220
 - b. 170
 - c. 270
 - d. 330
- 38** A number increased by 60% equals 416. Work backward from the answer choices to find the original number.
- a. 416
 - b. 320
 - c. 200
 - d. 260

- 39** A number increased by 15% equals 138. Work backward from the answer choices to find the original number.
- a.** 120
 - b.** 138
 - c.** 105
 - d.** 135
- 40** A number increased by 45% equals 116. Work backward from the answer choices to find the original number.
- a.** 125
 - b.** 80
 - c.** 116
 - d.** 35
- 41** A number increased by 25% equals 125. Work backward from the answer choices to find the original number.
- a.** 75
 - b.** 100
 - c.** 110
 - d.** 125
- 42** A number increased by 25% equals 75. Work backward from the answer choices to find the original number.
- a.** 60
 - b.** 35
 - c.** 85
 - d.** 75

- 43 A number increased by 5% equals 126. Work backward from the answer choices to find the original number.
- a. 115
 - b. 126
 - c. 120
 - d. 125
-

Multiple choice — hard

- 44 Which value of x satisfies $5x - 4 = 4x - 6$? Test the answer choices by substitution instead of solving algebraically.
- a. -3
 - b. 0
 - c. -1
 - d. -2
- 45 Which value of x satisfies $2x + 24 = 5x + 9$? Test the answer choices by substitution instead of solving algebraically.
- a. 5
 - b. 6
 - c. 4
 - d. 7

- 46 Which value of x satisfies $5x + 7 = 4x + 5$? Test the answer choices by substitution instead of solving algebraically.
- a. -1
 - b. -3
 - c. 0
 - d. -2
- 47 Which value of x satisfies $3x + 15 = 2x + 6$? Test the answer choices by substitution instead of solving algebraically.
- a. -10
 - b. -9
 - c. -8
 - d. -7
- 48 Which value of x satisfies $3x + 10 = 4x + 7$? Test the answer choices by substitution instead of solving algebraically.
- a. 2
 - b. 4
 - c. 5
 - d. 3
- 49 How many real solutions does $3x^2 - 3x + 2 = 0$ have? Use the discriminant to decide without graphing.
- a. 1
 - b. 2
 - c. 3
 - d. 0

- 50 How many real solutions does $x^2 - 3x + 3 = 0$ have? Use the discriminant to decide without graphing.
- a. 2
 - b. 1
 - c. 0
 - d. 3
- 51 How many real solutions does $4x^2 - 2x - 7 = 0$ have? Use the discriminant to decide without graphing.
- a. 2
 - b. 3
 - c. 1
 - d. 0
- 52 How many real solutions does $4x^2 + 7x = 0$ have? Use the discriminant to decide without graphing.
- a. 1
 - b. 2
 - c. 0
 - d. 3
- 53 How many real solutions does $2x^2 + 8x - 5 = 0$ have? Use the discriminant to decide without graphing.
- a. 2
 - b. 1
 - c. 3
 - d. 0

- 54 A calculator shows $\sqrt{648} \approx 25.456$, but the question asks for the exact answer in simplest radical form. Find it.
- a. $18\sqrt{3}$
 - b. $\sqrt{648}$
 - c. $18\sqrt{2}$
 - d. $19\sqrt{2}$
- 55 A calculator shows $\sqrt{117} \approx 10.817$, but the question asks for the exact answer in simplest radical form. Find it.
- a. $3\sqrt{13}$
 - b. $3\sqrt{14}$
 - c. $4\sqrt{13}$
 - d. $\sqrt{117}$
- 56 A calculator shows $\sqrt{936} \approx 30.594$, but the question asks for the exact answer in simplest radical form. Find it.
- a. $6\sqrt{26}$
 - b. $7\sqrt{26}$
 - c. $6\sqrt{27}$
 - d. $\sqrt{936}$
- 57 A calculator shows $\sqrt{128} \approx 11.314$, but the question asks for the exact answer in simplest radical form. Find it.
- a. $8\sqrt{2}$
 - b. $9\sqrt{2}$
 - c. $\sqrt{128}$
 - d. $8\sqrt{3}$

- 58 A calculator shows $\sqrt{608} \approx 24.658$, but the question asks for the exact answer in simplest radical form. Find it.
- a. $5\sqrt{38}$
 - b. $\sqrt{608}$
 - c. $4\sqrt{38}$
 - d. $4\sqrt{39}$